

Supplementary methods and results

Analysis provenance and missing inputs

This supplement accompanies the participant-level Weibo reanalysis. The original supplied study is known to have been published, but its article, abstract and results were not retrieved or used. Data collection details are attributed to the supplied Method. Original inputs and hashes, row-level inclusion decisions, all parameter candidates, warnings and failed runs, fitted models, predictions, resampling code, figures, and source documents are retained in the package.

The questionnaire transformation cannot be identified from decimal values alone. Original PANAS/SWLS unit conversions, item reliability, odd/even-text reliability, word clouds, demographic validity and temporal cutoff reconstruction were not performed. Metadata indicate text extending beyond questionnaire completion for 1,027 primary participants, so these results cannot establish prospective prediction. An ind value above one occurred for 34 participants, and 13 feature codes lack mappings. No post hoc distribution-based cleaning was applied.

Exact model selections and search records

The development-only grid had 42 configurations and 210 composite fold fits. Independent outcomes selected separate parameters; shared models selected one setting. All fitted transforms used only each fold's training rows. Candidate checkpoints preserve scores, warnings, status, and OOF arrays with a separate explicit UID row-order file. The initial vector-scaling preflight and corrected scalar-scaling final implementation are both retained. The correction was triggered by a synthetic numeric consistency check before holdout evaluation, with no change in grid or participant split.

Family	Selected parameters	CV normalized RMSE
Training mean	{ }	1.0016
Ridge	[{"alpha": 1000.0}, {"alpha": 10.0}, {"alpha": 1.0}]	0.9969
Independent RF	[{"max_features": 1.0, "min_samples_leaf": 15}, {"max_features": 0.5, "min_samples_leaf": 15}, {"max_features": 1.0, "min_samples_leaf": 15}]	0.9965
Independent ET	[{"max_features": 0.5, "min_samples_leaf": 15}, {"max_features": 1.0, "min_samples_leaf": 15}, {"max_features": 1.0, "min_samples_leaf": 15}]	0.9968
Independent HGB	[{"l2_regularization": 10.0, "max_leaf_nodes": 15}, {"l2_regularization": 1.0, "max_leaf_nodes": 31}, {"l2_regularization": 10.0, "max_leaf_nodes": 31}]	1.0226
Shared RF	{"max_features": 1.0, "min_samples_leaf": 15}	0.9963
Shared ET	{"max_features": 0.5, "min_samples_leaf": 15}	0.9977
Multitask elastic net	{"alpha": 0.01, "l1_ratio": 0.8}	0.9972

Note. All forests use 256 trees. RF uses bootstrapped trees; ET uses its default full training sample. HGB uses max_iter = 150, learning_rate = .05, min_samples_leaf = 20, early_stopping = False. Elastic net uses max_iter = 10000 and tol

= 1e-5. Ridge/independent ensemble parameters are listed in the order negative affect, positive affect, life satisfaction. Exact installed-library defaults are preserved in fitted models and environment/requirements.lock.txt.

Repeated development cross-validation

Three shuffled five-fold repetitions with seeds 20261023, 20261024 and 20261025 refitted the frozen selected configurations. Hyperparameters were not retuned. Fold variability is descriptive, not a confidence interval for unbiased performance; the same development sample had contributed to parameter selection.

Model	Outcome	Mean normalized RMSE	Across-fold SD
Shared RF	Negative affect	0.9993	0.0627
Shared RF	Positive affect	0.9995	0.0523
Shared RF	Life satisfaction	0.9923	0.0578
Independent RF	Negative affect	1.0034	0.0647
Independent RF	Positive affect	1.0003	0.0548
Independent RF	Life satisfaction	0.9899	0.0615
Training mean	Negative affect	1.0005	0.0647
Training mean	Positive affect	1.0011	0.0568
Training mean	Life satisfaction	1.0007	0.0570

Complete holdout performance

All families used the same 104 participants, independent of model selection. Pearson P values below are unadjusted and descriptive for nonprimary families. The primary winner's three outcome P values are Holm-adjusted in the main Results. Correlations for constant predictions are undefined. CSV-unit RMSE intervals condition on fitted models and the selected training/holdout partition.

Table S1 All model families on the primary holdout

Model	Outcome	Pearson r [95% CI]	p	RMSE [95% CI]
Training mean	Negative affect	Undefined	—	0.169 [0.150, 0.190]
Training mean	Positive affect	Undefined	—	0.170 [0.146, 0.192]
Training mean	Life satisfaction	Undefined	—	0.205 [0.184, 0.225]
Ridge	Negative affect	0.274 [0.086, 0.443]	0.005	0.169 [0.148, 0.189]
Ridge	Positive affect	0.330 [0.147, 0.492]	< .001	0.163 [0.140, 0.186]
Ridge	Life satisfaction	0.166 [-0.027, 0.348]	0.092	0.202 [0.181, 0.222]
Independent RF	Negative affect	0.230 [0.039, 0.405]	0.019	0.165 [0.144, 0.185]
Independent RF	Positive affect	0.325 [0.141, 0.487]	< .001	0.163 [0.141, 0.186]
Independent RF	Life satisfaction	0.184 [-0.008, 0.364]	0.061	0.201 [0.180, 0.222]
Independent ET	Negative affect	0.276 [0.088, 0.445]	0.005	0.165 [0.145, 0.186]
Independent ET	Positive affect	0.275 [0.087, 0.444]	0.005	0.165 [0.142, 0.187]
Independent ET	Life satisfaction	0.225 [0.033, 0.400]	0.022	0.200 [0.179, 0.220]
Independent HGB	Negative affect	0.075 [-0.119, 0.264]	0.447	0.172 [0.150, 0.194]
Independent HGB	Positive affect	0.219 [0.028, 0.395]	0.025	0.167 [0.146, 0.187]
Independent HGB	Life satisfaction	0.190 [-0.003, 0.369]	0.054	0.204 [0.180, 0.227]
Shared RF	Negative affect	0.247 [0.057, 0.419]	0.012	0.166 [0.146, 0.186]
Shared RF	Positive affect	0.284 [0.097, 0.452]	0.003	0.165 [0.143, 0.187]

Model	Outcome	Pearson r [95% CI]	p	RMSE [95% CI]
Shared RF	Life satisfaction	0.152 [-0.042, 0.335]	0.124	0.202 [0.181, 0.222]
Shared ET	Negative affect	0.312 [0.127, 0.476]	0.001	0.166 [0.145, 0.186]
Shared ET	Positive affect	0.306 [0.120, 0.471]	0.002	0.166 [0.142, 0.188]
Shared ET	Life satisfaction	0.227 [0.036, 0.402]	0.021	0.201 [0.180, 0.222]
Multitask elastic net	Negative affect	0.331 [0.148, 0.493]	< .001	0.166 [0.146, 0.186]
Multitask elastic net	Positive affect	0.402 [0.227, 0.552]	< .001	0.165 [0.143, 0.187]
Multitask elastic net	Life satisfaction	0.168 [-0.025, 0.350]	0.087	0.202 [0.182, 0.223]

Frozen-parameter sensitivity analyses

The latest-record analysis retains the same 1,037 participants and partitions, using the numerically latest questionnaire time. The singleton-ID analysis excludes all repeated IDs, reducing sample size. Restricted-feature analysis retains participants but omits ind and A, B, C1–C10, D, leaving 106 predictors. Every sensitivity refits the original chosen hyperparameters for each family without retuning. All family-by-scenario results, including R-squared and Spearman correlation, are retained in tables/tableS2_sensitivity.csv. The table below emphasizes the original winner; it does not select a new winner from sensitivity performance.

Table S2 Original winner across prespecified datasets

Dataset	Outcome	Test n	r	RMSE [95% CI]	R-squared
primary	Negative affect	104	0.247	0.166 [0.146, 0.186]	0.020
primary	Positive affect	104	0.284	0.165 [0.143, 0.187]	0.040
primary	Life satisfaction	104	0.152	0.202 [0.181, 0.222]	0.015
latest record	Negative affect	104	0.259	0.174 [0.154, 0.194]	0.024
latest record	Positive affect	104	0.286	0.163 [0.144, 0.182]	0.048
latest record	Life satisfaction	104	0.101	0.207 [0.185, 0.228]	0.008
singleton uids	Negative affect	85	0.303	0.171 [0.147, 0.195]	0.025
singleton uids	Positive affect	85	0.301	0.161 [0.138, 0.184]	0.055
singleton uids	Life satisfaction	85	0.173	0.207 [0.185, 0.229]	0.027
restricted features	Negative affect	104	0.301	0.165 [0.146, 0.185]	0.031
restricted features	Positive affect	104	0.268	0.165 [0.142, 0.188]	0.035
restricted features	Life satisfaction	104	0.125	0.203 [0.182, 0.223]	0.008

Attribution and permutation diagnostics

TreeSHAP used Shared RF fitted on 933 participants. The largest error in reconstructing training predictions from expected value plus feature attributions was 2.22e-16 CSV units. The union of the ten highest-ranked training features for each target contained 14 distinct features. These features were permuted one at a time on the 104 holdout participants with 20 shuffles each. Mean delta RMSE and shuffle SD measure conditional model sensitivity. Their uncertainty is not a population confidence interval, and correlated categories may substitute for each other. Full signed contributions, expected values, input values and participant keys are retained in results/primary/interpretation/training_shap_values.npz.

Outcome	Training-selected feature	Mean delta RMSE	Shuffle SD
Negative affect	Sexual	0.00008	0.00052
Negative affect	C10	0.00032	0.00050
Negative affect	C2	0.00025	0.00044
Negative affect	D	0.00007	0.00038
Negative affect	Angery	0.00011	0.00029
Positive affect	Sexual	0.00000	0.00054
Positive affect	We	0.00023	0.00041
Positive affect	Family	-0.00008	0.00034
Positive affect	D	0.00022	0.00030
Positive affect	Angery	0.00023	0.00030
Life satisfaction	Family	0.00103	0.00069
Life satisfaction	We	-0.00093	0.00074
Life satisfaction	C10	0.00006	0.00063
Life satisfaction	Sexual	-0.00017	0.00066
Life satisfaction	D	0.00093	0.00058

Independent computational reproduction

The final analysis was rerun from data.csv in a separate output directory using the locked environment. Participant manifests, CV row order, selected parameters, CV fold scores, primary predictions and reported primary metrics agreed within 1e-12. Maximum absolute primary prediction difference was 0.0e+00. This is computational reproducibility on the same input and environment, not an independent sample replication. The JSON verification report lists exact checks and any intentionally omitted nonprimary scenarios. Repeated development CV was retained from the original final run and was not necessary to repeat for the primary prediction equivalence check.

Data and document verification

The final main-paper bibliography contains 30 scholarly works, all published before 1 August 2024. All are stored in Zotero collection S2S9AGIR with readable full-text attachments and recorded source locations. The 16 additions were imported and read back through Zotero MCP; all imported local files matched retained source SHA-256 values. The collection contains 31 source attachments in total, including the prior Lundberg text/preprint pair. Diener et al. (2018) is retained as a text-only NCBI author-manuscript RTF; inline citation callouts and figure graphics are incomplete. The scanned Diener et al. (1999) source has local OCR for passage checks. Version distinctions and limitations are explicit in the evidence library. The target published study was not sought, acquired, used, or cited. Journal rules were checked against official sources on 23 September 2026. Author-controlled declarations and responsible human review remain pending.

Exploratory optimization after the primary analysis

The 22-configuration search was fixed after the original holdout had been inspected. It used the same 933 development participants and 104 holdout participants. Five outer development folds each performed five-fold inner selection, followed by a five-fold search on all development data: 132 candidate-by-search records

and 660 composite fold evaluations. All completed without failures or captured warnings. Scalar-target estimators were fitted independently unless the candidate explicitly shared outputs. Ridge, support-vector regression, elastic net and the shallow boosting alternatives tested different smoothness/regularization assumptions; PLS tested low-dimensional covariation. Every fitted imputer, scaler and target transformation used only the relevant training fold. Fixed log1p transformation was applied only to nonnegative input features. This was an exploratory extension, not a newly independent confirmation set.

Outer folds selected the original shared forest once, elastic net with alpha 0.1 and L1 ratio 0.5 twice, and independent forests with leaf size 60 twice. Variation in these choices is retained as a diagnostic. Table S3 gives unweighted means and descriptive SDs across the five outer folds; these SDs are not standard errors or confidence intervals, and overlapping training sets are not treated as independent replicates.

Table S3 Nested development validation

Procedure	Outcome	Mean normalized RMSE	Fold SD
Selected procedure	Negative affect	1.0044	0.0481
Selected procedure	Positive affect	0.9997	0.0822
Selected procedure	Life satisfaction	0.9976	0.0572
Original shared RF	Negative affect	1.0027	0.0473
Original shared RF	Positive affect	0.9986	0.0857
Original shared RF	Life satisfaction	0.9991	0.0578
Mean baseline	Negative affect	1.0020	0.0463
Mean baseline	Positive affect	1.0002	0.0836
Mean baseline	Life satisfaction	1.0001	0.0605

Note. Five outer development folds; all 933 development participants receive one outer-fold prediction per procedure. Fold scores divide outcome RMSE by that outer training fold's outcome SD. The selected procedure repeats candidate selection inside each outer training set. The other two comparators use fixed settings.

Table S4 Complete exploratory configuration search

Candidate	Parameters	Development CV score
mean	Fixed comparator	1.0007
original_rf	Fixed comparator	1.0001
ridge_raw_10	{"log": false, "alpha": 10}	1.0703
ridge_raw_100	{"log": false, "alpha": 100}	1.0298
ridge_raw_1000	{"log": false, "alpha": 1000}	1.0013
ridge_log_10	{"log": true, "alpha": 10}	1.0709
ridge_log_100	{"log": true, "alpha": 100}	1.0301
ridge_log_1000	{"log": true, "alpha": 1000}	1.0013
svr_0.1	{"C": 0.1}	1.0040
svr_1	{"C": 1}	1.0215
svr_10	{"C": 10}	1.1019
hgb_10	{"l2": 10}	1.0139
hgb_100	{"l2": 100}	1.0013
en_0.01_0.1	{"alpha": 0.01, "l1": 0.1}	1.0654
en_0.01_0.5	{"alpha": 0.01, "l1": 0.5}	1.0446
en_0.1_0.1	{"alpha": 0.1, "l1": 0.1}	1.0173
en_0.1_0.5	{"alpha": 0.1, "l1": 0.5}	0.9984

Candidate	Parameters	Development CV score
pls_2	{"components": 2}	1.0056
pls_5	{"components": 5}	1.0271
pls_10	{"components": 10}	1.0460
rf_30	{"leaf": 30}	0.9979
rf_60	{"leaf": 60}	0.9969

Note. All configurations use the same five full-development folds. Scores are macro normalized RMSE, averaged equally across folds and outcomes. Lower is better. RF models have 256 trees; the original shared RF retains minimum leaf size 15 and all features. The new independent RF models use half of the features per split. SVR uses an RBF kernel, gamma = scale, and epsilon = 0.1. HGB uses 7 leaves, 100 iterations, learning rate 0.05, and no early stopping. Elastic net uses 10,000 maximum iterations and tolerance 0.000001. PLS uses fold-local internal scaling. These scores select the final candidate and are not unbiased performance estimates of the search.

Table S5 Exploratory selected model on the previously seen holdout

Outcome	Pearson r [95% CI]	RMSE [95% CI]	R-squared
Negative affect	0.245 [0.055, 0.417]	0.1661 [0.1461, 0.1871]	0.0200
Positive affect	0.332 [0.149, 0.493]	0.1645 [0.1415, 0.1886]	0.0447
Life satisfaction	0.164 [-0.029, 0.346]	0.2023 [0.1821, 0.2223]	0.0155

Note. n = 104; independent random forests, leaf size 60, 256 trees, half of the features per split. Pearson intervals use Fisher z; RMSE intervals use 5,000 participant bootstraps. Units remain stored CSV units. The holdout had already informed earlier interpretation, so these are exploratory conditional intervals, not independent validation of the revised research process.

Table S6 Exploratory paired error comparisons

Comparison	Outcome	Delta RMSE	95% CI	p (Holm)
New vs mean	Negative affect	-0.00341	[-0.00670, -0.00038]	0.168
New vs mean	Positive affect	-0.00516	[-0.00819, -0.00215]	0.011
New vs mean	Life satisfaction	-0.00252	[-0.00643, 0.00137]	0.817
New vs original RF	Negative affect	-0.00002	[-0.00225, 0.00223]	1.000
New vs original RF	Positive affect	-0.00040	[-0.00254, 0.00161]	1.000
New vs original RF	Life satisfaction	-0.00009	[-0.00291, 0.00266]	1.000

Note. n = 104. Differences are the new selected model minus its comparator. Negative values favor the new model. Intervals use 5,000 paired participant bootstraps; P values use 10,000 prediction swaps followed by Holm adjustment over these six exploratory comparisons. All intervals for differences from the original forest include zero. The positive-affect mean-baseline comparison meets this within-extension adjustment, but does not resolve the exploratory status or absence of nested development improvement. Raw files retain the original comparison-function labels; model_a and model_b identify the actual procedures. A separate refit from raw data reproduced all revised predictions exactly (maximum absolute difference 0).

Supplementary figures

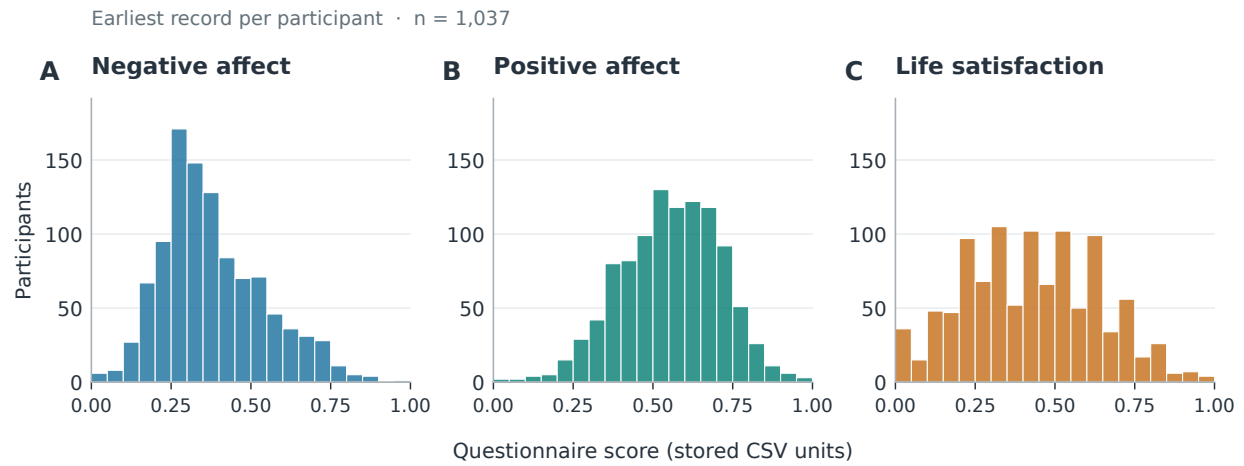


Figure S1. Primary score distributions for 1,037 participants using their earliest retained record. These are stored CSV score units; no inverse scaling is assumed. Bin widths are .05 and counts are participant counts.

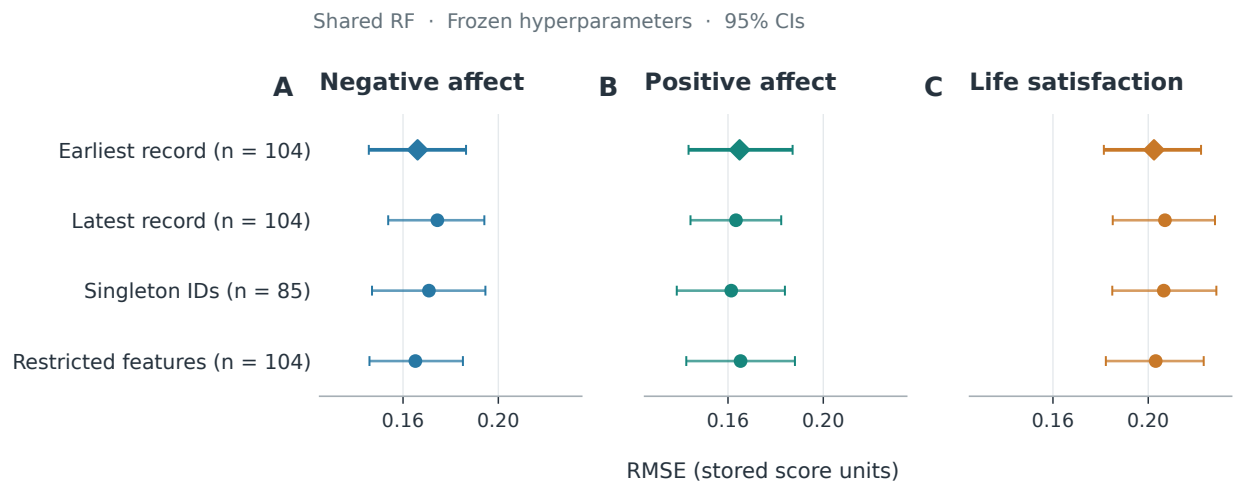


Figure S2. Original development-selected winner across the primary and three prespecified sensitivity datasets. Points are RMSE and bars are conditional 95% participant-bootstrap intervals in CSV units. Holdout sample sizes are provided in Table S2. Models are refitted with frozen primary parameters; the plot is not a new model-selection analysis.